

CHAPTER 2

In this chapter, you will study mixtures and their properties and characteristics. Here are some essential questions that you should be able to answer by the end of this chapter:

- What are mixtures and their different properties and characteristics?
- How do you classify mixtures?
- Why are mixtures important?
- Is it possible to live without mixtures?
- How does your knowledge of mixtures help you become wise consumers?
- What roles do suspensions, solutions, and colloids play in your everyday life?

Lesson 1

Properties of Mixtures

Do List

1. Investigate the properties of mixtures.
2. Demonstrate understanding of the characteristics of mixtures.

Energize Your Mind

Do you like milk tea? They come in various flavors and sizes. Milk tea is made by mixing tea and milk. Other ingredients such as sugar, honey, tapioca pearls, or other flavorings are added to satisfy and refresh those who enjoy it.

Milk tea is just one of the mixtures that you see and taste every day. Can you give other examples of mixtures? Perform the following activity to find out more about mixtures.

Explore Around

EVERYDAY MIXTURES

What You Need

pen

What to Do

Think about and answer the following questions. Write your answers in the space provided and share these with a partner.

1. What do you usually eat and drink for breakfast?
2. What art-related activities do you enjoy doing?
3. What other activities in school or at home require you to mix things?

What You Have Learned

Answer the following questions.

Write your answers in the space provided.

1. While eating your breakfast, do you notice that what you eat and drink are mixtures of different substances? What are these substances?
2. Do you mix things when you do art activities such as painting? What are the things that you mix?
3. What mixtures are involved in the other activities that you do in school or at home?

How You Learned

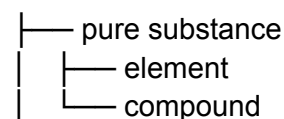
1. This activity required you to recall actions and things you have done. Was it easy for you to recall events and activities?
2. How did you determine that the common things that you use every day are mixtures?

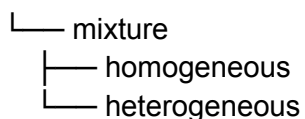
Equip Yourself

Study the chart below. See where mixtures fall under the classifications of matter.

[CHART: Classifications of Matter]

MATTER





classifications of matter

Matter is classified as a pure substance or a mixture. A substance is made up of atoms and molecules. It is a form of matter that possesses specific properties. Substances cannot be separated into components without breaking chemical bonds. Substances can be solids, liquids, gases, or plasma. When there are changes in temperature or pressure, substances shift between the different states of matter.

There are two types of substances—elements and compounds. An element is a substance that is made up of a particular kind of atom; thus, it cannot be broken down or transformed into another element by chemical means. An example of an element is carbon, which has only two carbon atoms. However, combining elements can cause a chemical reaction by which atoms are bonded together in molecules, creating a new substance called a compound. Water is an example of a compound because it is composed of the two elements: hydrogen and oxygen. Salt is a combination of the elements sodium and chlorine and is therefore a compound.



Substances are called "pure" to set them apart from mixtures. An example is pure water, which has the same properties and the same ratio of hydrogen to oxygen whether it is found in a river, prepared in a laboratory, or distilled mechanically. This is different from tap water which is a mixture formed by small amounts of chlorine and other minerals and seawater which contains compounds and ions. Other examples of substances in pure form include gold, diamond (carbon), and sugar (sucrose).

Mixtures

When two or more substances are physically combined, the result is a mixture. Salt combined with water forms a mixture called salt solution. A chocolate drink is a mixture made up of water, milk, sugar, and cocoa.

Mixtures are everywhere. The previous activity showed that you encounter mixtures every day, from the air you breathe and the food you eat to the things you use in your daily activities. Nature is abundant in mixtures. Look at the soil, rocks, and bodies of water. They are all a mixture of different substances.

General Properties of Mixtures

Mixtures have three general properties.

The components of the mixture retain their original properties. When substances combine to form a mixture, there is no chemical reaction involved. Thus, there is no new substance formed, and the properties of individual components remain the same. Mixtures combine physically with no definite proportions. You can regulate the amount of sugar in your milk tea drink. Manufacturers can determine the strength of steel bars by the predetermined quantities of iron and carbon. Fluoride in appropriate amounts may be added to drinking water in areas where there is a high incidence of dental problems.

The components of a mixture can easily be separated. Since mixtures are combined physically, they can be separated by mechanical means such as filtration and decantation. Filtration is a process by which solids are separated from liquids or gases by using a filter that allows the liquid through but not the solid. Decantation is a process by which mixtures are separated by removing a liquid layer that is free of a precipitate or a substance that is separated from a solution or suspension. Thus, a mixture of salt and water may be separated by evaporating the water, and salt is left as residue. This property of mixtures is in contrast to compounds that may be separated only by chemical means. Thus, the components of water, which are hydrogen and oxygen, can only be separated using a technique which involves a chemical reaction. Separation techniques will be discussed further in the next chapter.

Types of Mixtures

There are two types of mixtures—homogeneous and heterogeneous. A homogeneous mixture is made up of components that are equally distributed and blended together throughout it. Homogeneous mixtures can be solids, liquids, or gases. One example of a solid homogeneous mixture is cement mixed with sand, gravel and water to form concrete. A liquid homogeneous mixture is vinegar, which is made of water and acetic acid. An example of a gaseous homogeneous mixture is air. The air that you breathe is a mixture of oxygen, nitrogen, carbon dioxide, and other gases.

A heterogeneous mixture is a combination of substances that are not of uniform size and can be separated into its parts, which retain its original properties. A big part of the world is made up of

heterogeneous mixtures which may contain solid, liquid, or gaseous components. Examples of heterogeneous mixtures include fruit salad, cereals, mixed nuts, and many other kinds of food that we eat. Other examples are oil and water, the sand, stones, shells, and creatures on the beach, soil in your backyard, mud puddles, perfume, and smog.

The mixture of iron filings and sulfur shown in the picture is an example of a heterogeneous mixture. The iron filings and sulfur retain their properties even if they have been mixed. A magnet can be used to separate the iron filings from the mixture. This is possible because the iron filings are made of metal while the sulfur is not.

Look at your surroundings. Can you identify which of the things you see are compounds and which are mixtures? Do the following activity to find out.

Explore Around

MAGNET AND MIXTURE

What You Need

iron filings

wood carvings

Magnet

2 petri dishes or plastic saucers

iron supplement (capsule)

What to Do

1. Observe the iron filings and the wood carvings. Note their differences by writing their properties in the table below.

[TABLE]

Iron Filings | Wood Carvings

_____		_____
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2. Use the magnet to attract the iron filings. Use it also for the wood carvings. Observe the reaction of each material to the magnet.

3. Mix the iron filings with the wood carvings in the petri dish or plastic saucer. Observe the mixture.

4. Place the magnet near the mixture and see if it can still attract any of the substances.

5. Open an iron supplement capsule and pour out its contents on to another petri dish or plastic saucer.

Note: The iron in a supplement comes as a compound known as ferrous sulfate.

6. Place the magnet near the supplement powder and observe if it can attract any of its components.

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. What are the differences in the properties of iron and wood? Are these differences still observed or visible when you mixed them together?
2. What were the reactions of iron and wood to the magnet? Explain your answer.
3. What was the reaction of the iron supplement to the magnet?
4. How is the first setup (iron filings and wood carvings) different from the second (iron supplement powder)?

How You Learned

1. Which part of the activity did you find difficult? Why?
2. How will you make the activity easier to do?

From the activity, you see that components in a mixture retain their properties. These components may be separated by mechanical means, such as the use of a magnet.

The discussions show that objects and things around us are present as an element, compound, or mixture. Elements cannot be broken down into a simpler substance. A compound is a chemical combination of elements that form molecular bonds that cannot easily be split. A mixture is a physical combination of substances that retain the individual properties of its components and are bonded in any proportion. Too often, however, there is difficulty and some confusion in determining whether a material is a compound or a mixture.

The following table will help you understand the difference.

[TABLE: Compounds vs. Mixtures]

Compounds:

- made up of elements
- chemical combination
- can be separated chemically
- has fixed physical properties
(Example: Pure water boils at 100 °C.)
- examples: water, salt, sugar

Mixtures:

- made up of molecules of elements and compounds
- physical combination
- can be separated physically
- has varying physical properties
(Example: A mixture of water and alcohol boils over a range of temperature.)
- examples: cereal and milk, mud, seawater

Extend Your Understanding (Know Zone)

Did you know that blood is a mixture? It can be separated by a machine, called a centrifuge, into its two main parts: plasma and formed elements.

[SUMMARY BOX]

- Matter is classified as a pure substance or a mixture.
- A substance is made up of atoms and molecules.
- There are two types of substances—elements and compounds.
- An element is a substance that is made up of a particular kind of atom; thus, it cannot be broken down or transformed into another element by chemical means.
- Combining elements can cause a chemical reaction in which atoms are bonded together in molecules, creating a new substance called a compound.
- When two or more substances are physically combined, the result is a mixture.
- Mixtures have three general properties:
 - The components of the mixture retain their original properties.
 - Mixtures combine physically with no definite proportions.
 - The components of a mixture can easily be separated.
- There are two types of mixtures—homogeneous and heterogeneous.
- A homogeneous mixture is made up of components that are equally distributed and blended together throughout it.
- A heterogeneous mixture is a combination of substances that are not of uniform size and can be separated into its parts, which retain its original properties.

Express Your Learning

Answer the following questions:

1. How are mixtures different from compounds?
2. Which of the illustrations of particles show a mixture? Explain your answer.

[ILLUSTRATION a] — shows groups of paired circles (dark + light) clustered together in sets

[ILLUSTRATION b] — shows individual dark and light circles scattered randomly and unevenly

3. An unknown matter is presented to you. What steps will you take to find out if it is a mixture or a compound?
4. How did you feel about the lesson? Explain your answer.

Lesson 2

Classifications and Uses of Mixtures

Do List

1. Demonstrate understanding of the classification of mixtures.
2. Investigate the classification of mixtures.
3. Prepare beneficial or useful mixtures.
4. Enumerate the uses of different mixtures.
5. Distinguish colloids from other types of mixtures.
6. Identify the uses of colloids in daily life.

Energize Your Mind

Have you eaten halo-halo before? Did you like it? Halo-halo is a favorite dessert in the Philippines. It is made by mixing together shaved ice and ingredients such as nata de coco, jellies, pinipig, sweetened banana, ube, langka, ice cream, and milk. Others prefer to add more ingredients, depending on their taste. Before you mix the ingredients, you can clearly see all of its components. You can distinguish one from the other. But once the ice melts and you start mixing it, some components become difficult to see. Like halo-halo, some mixtures have components that are distinguishable and some that are not. Perform the activity on the next page to better understand the classifications of mixtures.



Explore Around

CLASSIFICATION OF MIXTURES

What You Need

cucumber
tomato
lettuce
salad dressing
croutons
powdered juice
cold water
hot water
tea bag containing green tea
calamansi
sugar
knife
large bowl
chopping board
spoon

What to Do

1. Wash the cucumber, tomato, and lettuce.
2. Cut the cucumber and tomato into slices. Ask your teacher for assistance.
3. Separate the lettuce leaves. Then slice them.
4. Prepare a garden salad by mixing together and tossing the lettuce, cucumber, tomato, and salad dressing in a bowl. Top the mixture with some croutons.
5. Prepare a juice drink by mixing the powdered juice and cold water thoroughly.
6. In a cup of hot water, place the tea bag and remove it after about two (2) to three (3) minutes. Cut the calamansi in half and gently squeeze each half of the fruit, seeing to it that only the liquid is added to the tea mixture. Add sugar and stir until the sugar is dissolved.

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. Do you still recognize the ingredients of your garden salad after you have mixed them together? Why do you say so?
2. Are the ingredients in your tea still distinguishable once you have removed the tea bag? Why do you say so?

3. How would you describe the appearance of the mixtures that you have prepared? What are the similarities and differences between the garden salad mixture and the juice drink mixture? What are the similarities and differences between the juice drink and the tea drink mixtures?

How You Learned

1. Was the activity fun to do? Why do you say so?
2. Did you have a clearer understanding about mixtures after doing the activity? Explain your answer.

Equip Yourself

As mentioned in the previous lesson, mixtures have two types: heterogeneous or homogeneous. A heterogeneous mixture is made up of components that are identifiable. It is also called nonuniform mixture. The garden salad you have mixed earlier is an example of a heterogeneous mixture. The sizes of its particles—the lettuce, tomato, cucumber, crouton, and dressing—are big enough to be distinguishable. More examples of heterogeneous mixtures are shown below. Can you identify the components of each mixture?

[IMAGES: heterogeneous mixtures — garden salad in a bowl, and another food mixture]



Unlike a heterogeneous mixture, a homogeneous mixture is one that has the same appearance all throughout. It is also called a uniform mixture. Although a homogeneous mixture is composed of different substances, its components are not distinguishable from one another and are spread out evenly as one phase. Phase is a distinctive form of matter that has the same chemical and physical properties throughout. The juice that you made is an example of a homogeneous mixture. The juice particles are uniformly distributed in water. These particles are too small that you could not see them with your naked eyes. This makes the mixture appear as one phase.

A mixture has two types: homogeneous (uniform) or heterogeneous (nonuniform). A homogeneous mixture appears as a single phase, whereas a heterogeneous mixture contains distinguishable components.

Extend Your Understanding (Know Zone)

An alloy is a mixture of elements that have metallic properties. One of the elements in an alloy is a metal. An example is steel, which is a mixture of iron and carbon.

Did you know that an alloy is a mixture of elements that have metallic properties? One of the elements in an alloy is a metal. An example is steel, which is a mixture of iron and carbon.

Classification of Mixtures

Mixtures are part of daily life. They are present in common items that are generally used in everyday activities. The classification of mixtures is determined by the size of the components or particles that are combined. A mixture can be a suspension, a solution, or a colloid.

Suspension

A suspension is a heterogeneous mixture with particles that are big enough to be seen with the naked eye, and are not evenly distributed throughout. It is created by combining or stirring two or more substances. When you mix a suspension, it might appear as if it has only one phase.

Many suspensions are a mixture of solid particles. When these mixtures are left for some time, the particles separate, with the bigger or heavier particles settling to the bottom. One example is cake mix, which has visible solid particles that are mixed together by stirring. Another example of a mixture of solids is soil.

Some suspensions are created when solid particles are mixed with a liquid or gas particles. When you mix a solid and a liquid by stirring or by shaking its container, and then leaving it for some time, you will observe that the particles separate, and the bigger and heavier particles settle at the bottom. For instance, if you mix sand with water, you will see that sand sinks to the bottom. This is also seen in medicine powder diluted with a prescribed amount of water. Another example of this kind of suspension is the mixture of smoke and dust particles that cause air pollution.

After a while, the solid particles settle to the ground as dust.

When the solid particles are lighter than a liquid, they will separate and the solid particles float to the top. This is seen when you mix water and baby powder. The baby powder floats to the top.

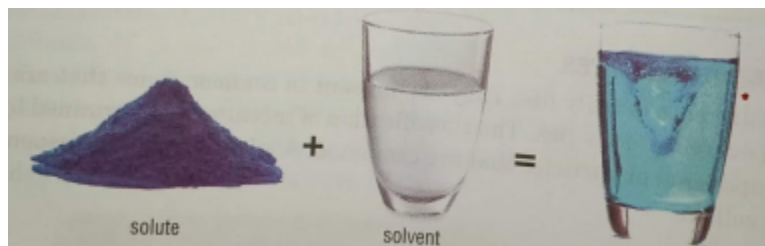
Solution

A solution is a homogeneous mixture in which one of the substances dissolves in the other. The substance that dissolves is called the solute. The substance in which the solute is dissolved is called the solvent. The solvent can be a solid, liquid, or a gas, and physical, not chemical process.

When the solute is put in a solvent, it undergoes dissolving. Dissolving is a process in which the solute, when placed in a solvent, breaks to an atomic or molecular level, and is no longer considered as an individual entity. For example, when a solute that is solid salt is placed in a

solvent which is liquid water, the salt dissolves in water and creates a saltwater solution. The salt breaks into the elements sodium and chlorine within the water solvent.

[DIAGRAM: solute + solvent = solution]



[CAPTION: components of a solution]

There are different types of solutions based on the kinds and amounts of solvents and solutes. Among these types are the aqueous or non-aqueous solutions and concentrated or diluted solutions. An aqueous solution is a solution in which any compound completely dissolves in water. A non-aqueous solution is a solution in which the solvent is not water but another kind of liquid such as oil or gas. A concentrated solution consists of large amounts of solute. A diluted solution has a large amount of solvent.

Some people are fond of drinking refreshing drinks like the ones that usually contain milk, fruit juice, tapioca pearls, and jellies.



[IMAGE: refreshing drink vendor with large containers of colorful drinks]
Photograph by junpinzon / Shutterstock.com

Colloid

A jelly is a special kind of mixture known as a colloid. A colloid falls in between a solution and a suspension in terms of properties. Do the next activity to discover how a colloid compares to a solution and a suspension.

Explore Around

IS THAT A COLLOID?

What You Need

salt solution

juice drink (pineapple or four seasons)

milk

clear glasses

small flashlight

What to Do

1. Transfer equal amounts of salt solution, juice drink, and milk in three separate clear glasses. To prepare a salt solution, combine half a teaspoon of salt with 200 mL of water. Mix thoroughly.
2. Leave the mixtures for about 15 minutes. Observe the three mixtures afterward.
3. Turn on the flashlight and point it at the salt solution. Observe.
4. Point also the flashlight at the juice drink and the milk. Observe.
5. Fill in the table below with your observations.

[TABLE]

Mixture	Physical Appearance	Effect of Light
salt solution	_____	_____
juice drink	_____	_____
milk	_____	_____

What You Have Learned

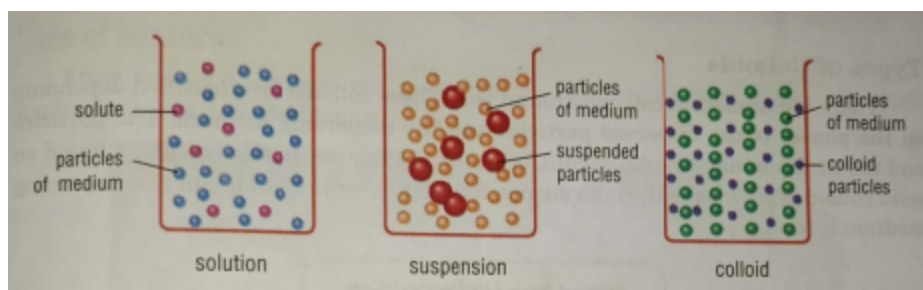
Answer the following questions. Write your answers in the space provided.

1. Which mixture(s) showed a visible streak of light? Why do you say so?
2. Which mixture(s) appeared as a single phase? Why do you say so?
3. Which of the mixtures do you think is a colloid? Why?

How You Learned

1. What problems did you encounter in this experiment?
2. How did you feel about these problems?
3. Do you think it is alright to encounter problems or errors in experiments? Why?

Study the illustration below. Can you see how the particles of a solution, suspension, and colloid differ from one another?



[DIAGRAM — particle sizes comparison]

A colloid is a special type of mixture. It appears as a single phase and is usually milky or opaque in appearance. It consists of slightly bigger particles than those in a solution. These particles are still too small to be seen by the naked eye. They remain suspended in a medium and do not settle like those in suspensions.

solution:

- solute
 - particles of medium
- (particles evenly and uniformly distributed, very small)

suspension:

- particles of medium
 - suspended particles
- (larger particles unevenly distributed)

colloid:

- particles of medium
 - colloid particles
- (medium-sized, evenly suspended but bigger than solution particles)

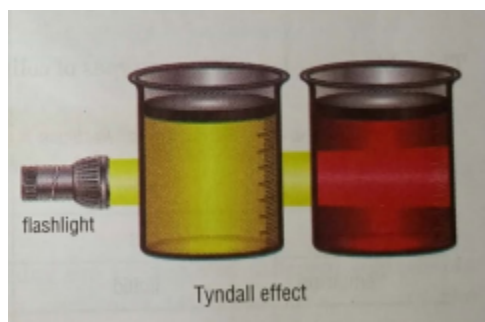
A colloid is a mixture that has particles bigger than those in a solution but smaller than those in a suspension.

Extend Your Understanding (Know Zone)

A colloid is a mixture that has particles bigger than those in a solution but smaller than those in a suspension.

Distinguishing a colloid from a suspension is easy because the particles of a suspension eventually separate. However, distinguishing a solution from a colloid is not as easy because they both appear as a single phase. A technique you can use to distinguish one from the other is to test for the Tyndall effect. The Tyndall effect is the ability of the particles of a colloid to scatter light that shines on them. A beam of light simply passes through a solution. On the other hand, a light beam flashed on a colloid is reflected and scattered by the larger particles and thus becomes more visible.

You actually tested for the Tyndall effect in the activity earlier. Do you recall how light is visible in milk but not in the salt solution? This tells you that milk is a colloid, and salt solution is not.



[DIAGRAM: flashlight → (left) suspension showing scattered light — Tyndall effect visible; (right) solution showing light passing through]

The Tyndall effect is the ability of particles of a colloid to scatter light that shines on them.

Types of Colloids

There are numerous colloids around you. These colloids are classified depending on the phases of the dispersed particles and the dispersing medium. The particles and the medium may be solid, liquid, or gas. For example, a jelly is a gel, a liquid on solid colloid. This means that the suspended particles are liquid, and the dispersing medium is solid.

Extend Your Understanding (Word Unlock)

To disperse means to "distribute or spread over an area." In a colloid, a dispersing medium refers to the substance that spreads the particles or where the particles are.

The table below lists different types of colloids.

[TABLE: Types of Colloids]

Type of Colloid	Dispersed Particles	Dispersing Medium	Example
sol	solid	liquid	paint
foam	gas	liquid	whipped cream
emulsion	liquid	liquid	mayonnaise
aerosol	solid/liquid	gas	smoke/fog
gel	solid	liquid	gelatin

Colloids are continuously being studied and enhanced. Scientists are optimistic that the science of colloids is moving toward endless possibilities. These studies will surely lead to more useful and powerful colloids in the future.

Colloids are classified according to the dispersed particles and the dispersing medium. Some types of colloids are foam, emulsion, and gel.

Uses of Mixtures

From nature to technology, mixtures have unlimited uses. Some important applications are described below.

Food

Most of the food you consume everyday are mixtures of different ingredients that are combined to obtain not only a particular taste but also to meet the nutritional needs of the body. Breads and pastries are mixtures that contain ingredients such as flour, yeast, and sugar. Viands are mixtures of vegetables, meat, and seasonings. Drinks are combinations of fruit, water, sugar, and flavors. Mayonnaise, a combination of oil and raw egg yolks, is a usual condiment paired with ketchup and mustard. Milk, butter, cheese, and other dairy products provide the calcium requirements of the body. Can you name other mixtures that are used as food?



[IMAGES: food mixtures — various Filipino dishes including noodle dish, sweet and sour dish, soup, and nuts/seeds]

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with ketchup and mustard. Milk, butter, cheese, and other dairy products provide the calcium requirements of the body. Can you name other mixtures that are used as food?

Medicine and Personal Care Products

Mixtures are used as medicines that help people maintain their health. When you are sick, you are sometimes given medicines that are mixtures of different compounds that combat organisms that cause sickness in the body. Daily supplements are composed of substances that help you meet your nutritional requirements and improve your resistance to diseases. In the hospital, patients may be given medications intravenously, that is, through the veins.

Colloids, in particular, are very useful means of storing and delivering a substance onto a target surface. Sunscreen lotion spreads the protective ultraviolet filters evenly over the skin. Colloidal calcium and gold are administered through injection for the treatment of certain diseases.



[IMAGE: medication given intravenously — IV drip in patient's hand]

Household Materials

Many of the materials you use at home are mixtures. Hygiene products like shampoo, toothpaste, and soap are usually compounds mixed with other substances such as fragrances and oils. Your clothes are textiles combined with paint and other decorative materials. Your gadgets and appliances are usually made of alloys and plastic. Furniture such as cabinets, tables, and sofa set are made with a combination of wood, fabrics, and metals.

Crude oil drawn from oil wells is often emulsified with water. Many insecticides are oil-in-water emulsions, thus, can easily be sprayed onto plants. The grease that is used to lubricate gears and machines is a water-in-oil emulsion. The foam used in fire extinguishers is made by mixing water, aluminum sulfate, and sodium bicarbonate with an absorptive agent. This foam can be used on all kinds of fire, including burning oils. Colloids like silica gel and alumina gel are used as adsorbents and desiccants. They keep food and other substances dry by reducing the humidity of surrounding air.



[IMAGE: silica gel used to keep food fresh — packets of silica desiccant]

Extend Your Understanding (Word Unlock)

- An adsorbent is a substance that is capable of holding molecules on its surface, forming a thin film.
- A desiccant is a material that absorbs moisture.

Mixtures in Nature

Things in nature are also mixtures. Tap water is a mixture of pure water and minerals. The air that you breathe is a mixture of different gases. The soil is a mixture of rocks, minerals, and decomposing minerals. The sky and the sea contain colloids in the form of dust particles and water impurities, respectively. The blue color of the sky and the sea is an effect of the scattering of light by these colloidal particles.

Extend Your Understanding (Eco-friendzee)

Smoke is a mixture of particles that are suspended in air. When these particles are suspended in and combine with fog, they form smog. Smog is a type of air pollution that is caused by burning coal and vehicular emissions. It is a serious environmental concern that may cause health problems to humans.

[SUMMARY BOX]

Mixtures have various uses. They are used as food, medicine, personal care products, and household materials. Mixtures are present in nature. Mixtures also have applications in research and technology.

Research and Technology

Colloids are used in many research and technological innovations. Colloids, particularly gels, are used to culture microorganisms. Agar-agar is commonly used to make such gels because it can stand high temperatures. Colloids are also used in producing artificial rain. Clouds are sprayed with charged colloidal particles to make the water combine and form large water droplets. Electronic paper or e-paper is another technology that uses colloids. E-paper is a thin and bendable display that mimics the appearance of ordinary ink on paper.

An e-paper is created by sandwiching colloids between two sheets of plastic. E-paper is currently used in e-books, displays, and cell phones.



[IMAGE: E-book using e-paper technology — purple e-reader device]

Extend Your Understanding (InfoTech)

E-paper makes it easy for people to read text and allow for a wider viewing angle than most light-emitting displays. That is because e-paper reflects light.

[SUMMARY BOX — Lesson 2 Key Points]

- A mixture can be categorized as homogeneous (uniform) or heterogeneous (nonuniform). A homogeneous mixture appears as a single phase, while a heterogeneous mixture contains distinguishable components.
- A mixture can be a suspension, solution, or colloid.
- A suspension is a heterogeneous mixture with particles that are big enough to be seen with the naked eye, and are not evenly distributed throughout.
- A solution is a homogeneous mixture in which one of the substances dissolves in the other. The substance that dissolves is called the solute. The substance in which the solute is dissolved is called the solvent.
- A colloid is a mixture that has particles bigger than those in a solution but smaller than those in a suspension.
- Mixtures have a variety of uses. They are used as food, medicine, personal care products, and household materials. Mixtures are present in nature. Mixtures also have applications in research and technology.

Express Your Learning

Answer the following questions:

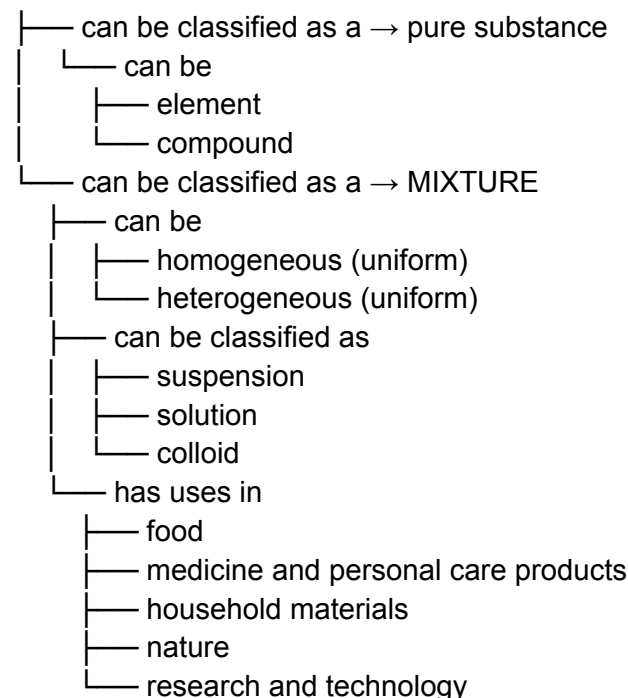
1. Differentiate homogeneous mixtures from heterogeneous mixtures.
2. You mixed sand with water. What kind of mixture did you make? Explain your answer.
3. Identify a mixture not mentioned in this chapter and explain its importance.

4. How did you feel about the lesson? Explain your answer.
5. How is a colloid different from a solution and a suspension?
6. Why do colloids exhibit the Tyndall effect?
7. In a range of 1 to 5 (5 being the highest), how would you rate the importance of colloids in the field of medicine? Explain your answer.

Map It!

[CONCEPT MAP]

matter



Wrap Up!

- Matter is classified as a pure substance or a mixture.
- A substance is made up of atoms and molecules.
- There are two types of substances—elements and compounds.
- An element is a substance that is made up of a particular kind of atom; thus, it cannot be broken down or transformed into another element by chemical means.
- Combining elements can cause a chemical reaction in which atoms are bonded together in molecules, creating a new substance called a compound.
- When two or more substances are physically combined, the result is a mixture.
- Mixtures have three general properties:
 - The components of a mixture retain their original properties.

- Mixtures combine physically, with no definite proportions.
- The components of a mixture can be easily separated.

[WRAP UP continued]

- A mixture can be categorized as homogeneous (uniform) or heterogeneous (nonuniform). A homogeneous mixture appears as a single phase, while a heterogeneous mixture contains distinguishable components.
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- A suspension is a heterogeneous mixture with particles that are big enough to be seen with the naked eye, and are not evenly distributed throughout.
- A solution is a homogeneous mixture in which one of the substances dissolves in the other. The substance that dissolves is called the solute. The substance in which the solute is dissolved is called the solvent.
- A colloid is a mixture that has particles bigger than those in a solution but smaller than those in a suspension.
- Mixtures have a variety of uses. They are used as food, medicine, personal care products, and household materials. Mixtures are present in nature. Mixtures also have applications in research and technology.

Size Up!

Knowledge

Encircle the letter of the correct answer.

1. Which describes a mixture?
 - a. a pure substance
 - b. a chemical reaction
 - c. a physical combination of substances
 - d. a chemical interaction of substances
2. Which of the following statements correctly describes a compound and a mixture?
 - a. A compound has one kind of atom, whereas a mixture has more than one kind of atom.
 - b. A compound can be separated chemically, whereas a mixture can be separated physically.
 - c. A compound is artificially made, whereas a mixture is naturally made.
 - d. A compound has big components, whereas a mixture has small components.
3. Which statement is true about the particles of colloids?
 - a. They scatter light.
 - b. They are big enough to settle.
 - c. They can be seen by the naked eyes.
 - d. They can be separated by chemical means.

4. A glass contains a mixture. Some substances in the mixture settle at the bottom of the glass after some time. What kind of mixture is in the glass?

- a. a solution
- b. a suspension
- c. a colloid
- d. a homogeneous mixture

5. What is observed in the Tyndall effect?

- a. the settling of particles
- b. the separation of substances
- c. an electric charge
- d. a beam of light

Process/Skill

Encircle the letter of the correct answer.

1. Which observation indicates that an unknown substance is a mixture?

- a. The substance is hot.
- b. There are different particles in the substance.
- c. The substance can conduct electricity.
- d. The substance is white.

2. Which material can help identify if a mixture is colloid?

- a. a battery
- b. a laser
- c. a metal
- d. a magnet

3. Your teacher asked you to arrange a colloid, a solution, and a suspension according to increasing particle size. Which shows the correct order?

- a. colloid–solution–suspension
- b. colloid–suspension–solution
- c. solution–suspension–colloid
- d. solution–colloid–suspension

4. You are given a solution and a suspension. What can you do to distinguish one from the other?

- a. Stir both mixtures and see the effect.
- b. Leave the mixtures and see which one will settle.
- c. Test for Tyndall effect.
- d. Drop a marble into each mixture and see how it reacts.

For numbers 5–7, choose your answers from the illustrations below. Each dot represents an atom.

[ILLUSTRATIONS — four boxes labeled I, II, III, IV]

- I — clusters of paired dark circles (same type of atom bonded together)
- II — groups of two different circles bonded in pairs (dark + light pairs)
- III — mixed dark and light circles scattered randomly (heterogeneous mixture)
- IV — mixed dark and light circles evenly spread (homogeneous mixture)

5. Which has particles that may be separated by physical means?

- a. I b. II c. III d. IV

6. Which is an element?

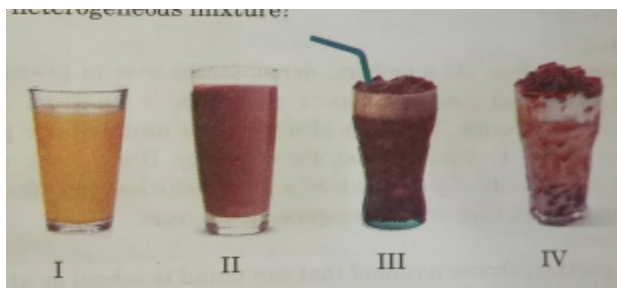
- a. I b. II c. III d. IV

7. Which is a mixture?

- a. I b. II c. III d. IV

8. Which is a heterogeneous mixture?

- a. I b. II c. III d. IV



[IMAGES: four drinks in glasses labeled I, II, III, IV — from lightest to darkest colored, IV has a red topping]

9. Below are observations made about an unknown mixture.

- uniform in appearance
- looks like a liquid
- does not scatter light

Which conclusion may be derived from these observations?

- a. The unknown mixture is heterogeneous.
- b. The unknown mixture is a colloid.
- c. The unknown mixture is a suspension.
- d. The unknown mixture is a solution.

10. A colloid is made by allowing nitrogen gas to be dispersed in water. What type of colloid is formed?

- a. sol b. aerosol c. foam d. gel

Understanding

Answer the following questions:

1. Using real-life examples, explain the characteristics of a mixture.
2. In what ways do you use mixtures in your everyday life?
3. Is it possible to live without mixtures? Why?

Product

Do the following tasks:

1. Delicious Mixtures

As a group of four (4) members, demonstrate how to prepare any food of your choice. The food you will choose should be a mixture. Before demonstrating, look for a recipe. You may also use your own or your parents' recipe. Present your recipe to your teacher for checking. During the demonstration, identify the components of your mixture and tell whether your final food product is a homogeneous or a heterogeneous mixture.

2. Colloid in Focus

With a partner, choose a colloid that can be found in school or at home. Present a report about the colloid and include a picture. Identify the type of the colloid and enumerate its uses. Make your PowerPoint presentation as creative and interesting as possible.

3. Mixture Booth

As a group, bring 10 mixtures and categorize them as homogeneous or heterogeneous. Classify them further into solutions, suspensions, and colloids. Explain to the class how you categorized the mixtures.

